# Introduction to Programming – Midterm Quiz 1

**Fall 2026-27** | Time: **40 minutes** | Marks: **30**

**Name:** _____ **ID:** 26 – _____ – 3 **Section**: _____

**Instructions.** Write a **complete C++ program** (including `main`) on the answer sheet. It takes **no input**. Use a **for loop**; you may use only `if / else if / else`, `switch`, arithmetic, relational and logical operators.

**Step 1:** work out your own variable values from your ID using the table below and write them in its last column. **Step 2:** declare them as variables (same names) in your program and solve the problem. You are marked on your **code**, not its output, so you need not work out the final numbers. *Write your code on the other side.*

## Step 1: your constants from your ID

Take the **last three digits of the number between the dashes** in your ID and call them a b c, left to right. For example, ID 26-66***345***-3 gives a = 3, b = 4, c = 5.

| Variable Name | Value | Example (a=3, b=4, c=5) | Calculate Your value |
|---|---|---|---|
| **N** | 30 + a | (30 + a) = 33 | _____ |
| **L1** | 5 + b | (5 + 4) = 9 | _____ |
| **L2** | L1 + c | (9 + 5) = 14 | _____ |
| **R1** | 20 + c | (20 + 5) = 25 | _____ |
| **R2** | R1 + 10 | (25+10) = 35 | _____ |
| **R3** | R1 + 20 | (25 + 20) = 45 | _____ |
| **D** | 3 + b | (3 + 4) = 7 | _____ |

## Parcel Charges at a Dhaka Courier Shop

A courier shop in Dhaka sends parcels to Chattogram. The **rate per kg depends on the weight band** the parcel falls in. One parcel of each weight arrives at the counter: **1 kg, 2 kg, 3 kg, ... up to N kg**. For every parcel, apply:

1. **Rate.** A parcel up to and including *9* kg costs *25* taka per kg. Above *L1 (10)* and up to and including *L2 14* kg it costs *35* taka per kg. Above *L2* kg it costs *R3* taka per kg.
2. **Charge.** Charge = weight x rate (the whole parcel uses its band's rate).
3. **Offer.** If the weight is a **multiple of *D***, the shop gives **20 Taka discount** on that parcel's charge.

Your code should print exactly two lines: the **total** amount to pay for all parcels, and **how many parcels got the offer**. (Replace `<number>` with the actual number you code calculates.)

```
Total charge: <number>
Discounted parcels: <number>
```

### An example

If, **N = 33, L1 = 9, L2 = 14, R1 = 25, R2 = 35, R3 = 45, D = 7**.

1. You have to calculate charge for each 1 … 33 (*N*) parcels.
2. Parcel weight 1kg to 9kg (*L1*): charge is 25 taka (R1) per kg
3. Parcel weight 10kg (*L1*+1) to 14kg (*L2*): charge is 35 taka (R2) per kg
4. Parcel weight 15kg (*L2+1*) to 33kg (N): charge is 45 taka (R3) per kg
5. If a parcel's weight is 14kg: charge will be 14x35 = 490 taka
6. In this case, D = 7, so if a parcel's weight can be divided by 7 (with remainder 0), the parcel will get 20 taka discount. If a parcel is 14kg, it is divisible by 7,

   Therefore the final charge after discount will be (14x35) – 20 = 470 taka
7. Keep a count of each parcel that got the discount.
8. You will have to calculate price for each parcel and sum them and print the total.
9. You will also have the print the count of how many parcels got the discount

*Write your code on the other side.*